

Project Constraints Essay

The development of an in-flight UAV powertrain health monitoring framework is shaped by several important constraints that influence both technical and non-technical design decisions. These constraints affect hardware selection, machine learning approach, testing procedures, and the overall scope of what can be achieved in the project. Addressing them carefully ensures that the final system remains feasible, safe, and aligned with real-world operational conditions.

Economic constraints are significant because this is an undergraduate project with limited funding. High-end microphones and specialized sensors such as those used at NASA by Dr. Kulkarni are not feasible to procure. Instead, cost-effective microphones and open-source software are being used, which impacts accuracy, scalability, and sensor fidelity. Professional constraints also play a role because the project involves advanced mechanical engineering, sensor integration, and machine learning. This requires careful planning, disciplined execution, and maintaining engineering standards that reflect well on all collaborators and the institutions involved.

Ethical considerations are critical because the system is designed to detect faults in aerial vehicles. In real-world operations, detecting a healthy system as faulty is preferable to missing a fault and allowing a UAV to fly in unsafe conditions. Faulty real-world data is limited, and supervised models would likely bias toward healthy data due to class imbalance. A semi-supervised approach is therefore used to build a more reliable system that prioritizes safety over raw performance.

Legal constraints are relevant because UAV testing must comply with federal and institutional safety regulations. This limits where and when flight testing can occur and imposes strict requirements on flight procedures. Security is not a major technical constraint since the collected data will be made public, but UAVs are often deployed in security-sensitive applications. This raises the importance of safe operation in environments where drones may face interference or hazardous conditions.

Environmental and social factors influence the project through its expected benefits. By enabling early fault detection and predictive maintenance, the project aims to reduce crash risks, minimize environmental debris, and support more reliable UAV deployment in applications such as inspection, logistics, and emergency response. Diversity and cultural impact are represented through the interdisciplinary structure of the team. Meeting both mechanical engineering and computer science requirements adds coordination complexity, but also strengthens the project by combining different areas of expertise.

These constraints collectively shape the system's design choices and research direction. By recognizing them early and building the project around realistic technical and operational boundaries, the team can deliver a solution that is safe, feasible, and impactful in real-world UAV applications.

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